

**ATTACHMENT E – MONITORING AND REPORTING PROGRAM NO. R1-2014-0002
(REVISED MAY 19, 2015)**

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

The Code of Federal Regulations (40 C.F.R. § 122.48) requires that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Regional Water Board to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements that implement federal and California regulations.

I. GENERAL MONITORING PROVISIONS

- A. Wastewater Monitoring Provision.** Composite samples may be taken by a proportional sampling device approved by the Executive Officer or by grab samples composited in proportion to flow. In compositing grab samples, the sampling interval shall not exceed one hour.
- B. Supplemental Monitoring Provision.** If the Permittee monitors any pollutant more frequently than required by this Order, using test procedures approved by 40 C.F.R. Part 136 or as specified in this Order, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the monthly and annual discharge monitoring reports.
- C. Data Quality Assurance Provision.** Laboratories analyzing monitoring samples shall be certified by the California Department of Public Health (CDPH), in accordance with the provision of Water Code section 13176, and must include quality assurance/quality control data with their reports.
- D. Instrumentation and Calibration Provision.** All monitoring instruments and devices used by the Permittee to fulfill the prescribed monitoring program shall be properly installed, calibrated, operated, and maintained to ensure that the accuracy of the measurements is consistent with the accepted capability of that type of device. All flow measurement and UV transmittance devices shall be calibrated no less than the manufacturer's recommended intervals or one year intervals, (whichever comes first) to ensure continued accuracy of the devices.
- E. Minimum Levels (ML) and Reporting Levels (RL).** Compliance and reasonable potential monitoring analyses shall be conducted using commercially available and reasonably achievable detection limits that are lower than the applicable effluent limitation. If no ML value is below the effluent limitations, the lowest ML shall be selected as the RL.

II. MONITORING LOCATIONS

The Permittee shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Table E-1. Monitoring Station Locations

Discharge/ Distribution Point Name	Monitoring Location Name	Monitoring Location Description
---	INF-001	Untreated influent wastewater collected at the plant headworks at a representative point preceding primary treatment. Latitude: 38.48535° N Longitude: 122.999° W
---	INT-001A	Location for monitoring surface loading rate through AWT filters.
---	INT-001B	Treated wastewater immediately following the AWT process for monitoring AWT turbidity and effluent flow.
---	INT-002	Location for monitoring ultraviolet (UV) radiation dose and UV transmittance of the UV disinfection system

Discharge/ Distribution Point Name	Monitoring Location Name	Monitoring Location Description
001	EFF-001	Treated wastewater after disinfection, but prior to storage (for monitoring technology-based effluent limitations). Latitude: 38.4854° N Longitude: 122.9982° W
002	EFF-002 ¹	Location following storage where representative sample of treated, disinfected effluent may be collected prior to discharge to Russian River (for monitoring WQBELs). Latitude: 38.482087° N Longitude: 123.000052° W
003	LND-001 ¹	Location where representative samples of treated wastewater, to be used for irrigation on the Burch property, can be collected, following all treatment and storage and immediately before its application for irrigation.
004	REC-001 ¹	Location where representative samples of treated wastewater, to be used for recycled water irrigation at the Northwood Golf Course or other approved recycled water use sites, can be collected, following all treatment and on-site storage and immediately before its application for irrigation.
---	RSW-001	Upstream receiving water monitoring location. Samples shall be representative of background conditions in the Russian River. Latitude: 38.4834° N Longitude: 123.0106° W
---	RSW-002	Downstream receiving water monitoring location. Samples shall be representative of conditions in the Russian River following introduction and mixing of effluent from the wastewater treatment Facility. Latitude: 38.4812° N Longitude: 122.9955° W
---	GW-001	Down-gradient monitoring well, located in the lower Burch irrigation area and approximately 225 feet down-gradient of monitoring well GW-002.
---	GW-002	Down-gradient monitoring well, located at the up-gradient edge of the lower Burch irrigation area.
---	GW-003	Up-gradient groundwater monitoring well. This well is located at the entrance to the Facility and is approximately 500 feet up-gradient of the lower Burch irrigation area.
---	BIO-001	A representative sample of the sludge or biosolids generated when removed for disposal.
Table Notes: 1. Monitoring Locations EFF-002, LND-001, and REC-001 may be the same location, the sampling tap following the on-site effluent storage pond. Unique sampling location names were given to differentiate the three different disposal methods which each have different monitoring requirements. Abbreviations: INF- Influent; INT- Internal; EFF- Effluent; RSW- Receiving Surface Water; REC- Reclamation; GW- Groundwater; BIO- Biosolids		

The North latitude and West longitude information in Table E-1 are approximate for administrative purposes.

III. INFLUENT MONITORING REQUIREMENTS

A. Monitoring Location INF-001

1. The Permittee shall monitor influent to the Facility at Monitoring Location INF-001 as follows:

Table E-2. Influent Monitoring, Monitoring Location INF-001

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Influent Flow ¹	mgd	Continuous	Daily	---
BOD ₅	mg/L	24-hour composite	Weekly	Standard Methods ²
TSS	mg/L	24-hour composite	Weekly	Standard Methods ²
Table Notes: 1. Each month, the Permittee shall report average daily and average monthly flows. 2. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or current test procedures specified in 40 C.F.R. Part 136.				

IV. EFFLUENT MONITORING REQUIREMENTS

A. Monitoring Location EFF-001, Discharge to Effluent Storage

1. The Permittee shall monitor treated wastewater at Monitoring Location EFF-001 as follows. If more than one analytical test method is listed for a given parameter, the Permittee must select from the listed methods and corresponding Minimum Level:

Table E-3. Effluent Monitoring – Monitoring Location EFF-001 – Discharge to Effluent Storage

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and (Minimum Level, units), respectively
Effluent Flow ^{1, 2}	mgd	Continuous	Daily	Meter
BOD ₅	mg/L	24-hour composite	Weekly	Standard Methods ³
	lbs/day	Calculate		
	Monthly % Removal	Calculate		
TSS	mg/L	24-hour composite	Weekly	Standard Methods ²
	lbs/day	Calculate		
	Monthly % Removal	Calculate		
pH	pH units	Grab	Daily	Standard Methods ²
Total Coliform Organisms ⁴	MPN/100 mL	Grab	Daily	Standard Methods ²
Radioactivity ⁵	pCi/L	Grab	1X/permit term	40 C.F.R. 136
Table Notes: 1. Treated effluent flow may be monitored at Monitoring Location INT-001B. 2. Each month, the Permittee shall report peak daily, average daily, and average monthly flows. 3. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or current test procedures specified in 40 C.F.R. Part 136. 4. Report daily test results and 7-day medians.				

5. Radionuclides measured shall include Combined Radium-226 and Radium 228, Gross Alpha, Gross Beta, Tritium, Strontium-90, and Uranium.

B. Monitoring Location EFF-002, Discharge to Russian River

1. The Permittee shall monitor treated wastewater to be discharged to the Russian River prior to contact with the receiving water at Monitoring Location EFF-002 as follows:

Table E-4. Effluent Monitoring – Monitoring Location EFF-002 – Discharge to Russian River

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and (Minimum Level, units), respectively
Effluent Flow ¹	mgd	Continuous	Daily	Meter
Dilution Rate	% of stream flow	Calculation	Daily	---
pH	pH units	Grab	Daily	Standard Methods ²
Dissolved Oxygen	mg/L	Grab	Daily	Standard Methods ²
Temperature	°C	Grab	Daily	Standard Methods ²
Aluminum	mg/L	Grab	Monthly	Standard Methods ²
Ammonia Nitrogen, Total (as N) ³	mg/L	Grab	Monthly	Standard Methods ²
Unionized Ammonia (as N)	mg/L	---	Monthly	Calculation
Nitrate Nitrogen, Total (as N)	mg/L	Grab	Monthly	Standard Methods ²
Organic Nitrogen, Total (as N)	mg/L	Grab	Monthly	Standard Methods ²
Phosphorus, Total (as P)	mg/L	Grab	Monthly	Standard Methods ²
Hardness	mg/L	Grab	Once per Discharge Season ⁴	Standard Methods ²
Copper	µg/L	Grab	Once per Discharge Season	
Bis (2-Ethylhexyl) Phthalate	µg/L	Grab	2X/dischage season	EPA Method 625
Acute Toxicity Bioassay – two species	Percent survival	Grab	Once per Discharge Season	See Section V.A
Chronic Toxicity Bioassay	TUc	Grab	Once per Discharge Season	See Section V.B
Chronic Toxicity (Narrative)	Passed/ Triggered ⁵			
CTR Priority Pollutants ^{6,7}	µg/L ⁸	Grab	1X per Permit Term	40 C.F.R. 136
Title 22 Pollutants ⁹	µg/L ⁸	Grab	1X per Permit Term	Standard Methods ²

Table Notes:

1. The Permittee shall report average daily, maximum daily, and average monthly flows.
2. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public

Health Administration) or current test procedures specified in 40 C.F.R. Part 136. 3. Monitoring for ammonia shall be concurrent with acute whole effluent toxicity monitoring (section V.A.1 of the MRP). Temperature and pH shall be recorded at the same time that the ammonia sample is collected. 4. Effluent hardness shall be monitored concurrently with the once per discharge season copper sample and the once per permit term priority pollutant sample. 5. The Permittee shall include reporting regarding compliance with the narrative toxicity objective in Receiving Water Limitation V.A.10 by reporting whether the chronic toxicity test "passed" or "triggered" in relation to the chronic toxicity trigger of 1.6 TUC (where TUC=100/NOEC) for each single sample or 1.0 TUC as a monthly median. For narrative chronic toxicity reporting, "Passed" shall be reported when chronic toxicity effluent results do not trigger accelerated testing (e.g., a single sample result of ≤ 1.6 TUC or a monthly median of ≤ 1.0 TUC). "Triggered" shall be reported when chronic toxicity effluent results trigger accelerated testing by exceeding the chronic toxicity trigger of 1.6 TUC for a single sample or 1.0 TUC as a monthly median. 6. Those pollutants identified as Compound Nos. 1 – 126 by the California Toxics Rule at 40 C.F.R. 131.38 (b) (1). Samples shall be collected on the same day as receiving water samples are collected for analysis of the priority pollutants. Analyses for the priority pollutants shall be conducted in accordance to methods established at 40 C.F.R. 136, or if no method is specified for a pollutant at 40 C.F.R. 136, in accordance to methods approved by the State Water Resources Control Board or the Regional Water Board. Holding times for unpreserved cyanide shall not exceed one hour. 7. Analytical methods shall achieve the minimum levels (ML) specified in Appendix 4 of the SIP. If no ML value is below the effluent limitation, the lowest ML shall be selected as the reporting level (RL). In accordance with section 2.4.1 of the SIP, the Permittee shall report the Reporting Level (RL) and the Method Detection Limit (MDL) with each sample result. 8. Or other appropriate units. 9. Title 22 Drinking Water Constituents are those pollutants for which the California Department of Public Health has established Maximum Contaminant Levels (MCLs) at title 22, division 4, chapter 15, article 4, section 64431 (Inorganic Chemicals) and article 5.5, section 64444 (Organic Chemicals) of the CCR. Duplicate analyses are not required for pollutants that are identified both as CTR and title 22 pollutants. Monitoring required in future permit terms may be reduced to only those pollutants detected in the title 22 sampling conducted during this term.	
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V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

A. Acute Toxicity Testing

The Permittee shall conduct acute whole effluent toxicity testing (WET) to determine compliance with the effluent limitation for acute toxicity established by section IV.A.2.b of the Order.

- 1. Test Frequency.** The Permittee shall conduct acute WET testing in accordance with the schedule established by this MRP while discharging at Discharge Point 002, as summarized in Table E-4, above.
- 2. Sample Type.** For 96-hour static renewal or 96-hour static non-renewal testing, the effluent samples shall be grab samples collected at Monitoring Location EFF-002.
- 3. Test Species.** Test species for acute WET testing shall be with an invertebrate, the water flea (*Ceriodaphnia dubia*) and a vertebrate, the rainbow trout (*Oncorhynchus mykiss*).
- 4. Test Methods.** The presence of acute toxicity shall be estimated as specified in *Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms* (U.S. EPA Report No. EPA-821-R-02-012, 5th edition or subsequent editions), or other methods approved by the Executive Officer.

Test procedures related to pH control, sample filtration, aeration, temperature control and sample dechlorination shall be performed in accordance with the U.S. EPA method and fully explained and justified in each acute toxicity report submitted to the Regional Water Board. The control of pH in acute toxicity tests is allowed, provided the test pH is maintained at the effluent pH measured at the time of sample collection, and the control of pH is done in a manner that has the least influence on the test water chemistry and on the toxicity of other pH sensitive materials such as some heavy metals, sulfide and cyanide.

5. **Test Dilutions.** The acute toxicity test shall be conducted using 100 percent effluent collected at Monitoring Location EFF-002.
6. **Test Failure.** If an acute toxicity test does not meet all test acceptability criteria, as specified in the test method, the Permittee shall re-sample and re-test as soon as possible, not to exceed 7 days following notification of test failure.
7. **Accelerated Monitoring.** If the result of any acute toxicity test fails to meet the single test minimum limitation (70 percent survival), and the testing meets all test acceptability criteria, the Permittee shall take two more samples, one within 14 days and one within 21 days following receipt of the initial sample result. If any one of the additional samples do not comply with the three sample median minimum limitation (90 percent survival), the Permittee shall initiate a Toxicity Reduction Evaluation (TRE) in accordance with section VI.C.2.a. of the Order. If the two additional samples are in compliance with the acute toxicity requirement and testing meets all test acceptability criteria, then a TRE will not be required. If the discharge stops before additional samples can be collected, the Permittee shall contact the Executive Officer within 21 days with a plan to demonstrate compliance with the effluent limitation.
8. **Notification.** The Permittee shall notify the Regional Water Board verbally within 72 hours and in writing 14 days after receipt of test results exceeding the acute toxicity effluent limitation during regular or accelerated monitoring. The notification shall describe actions the Permittee has taken or will take to investigate and correct the cause(s) of toxicity. It may also include a status report on any actions required by this Order, with a schedule for actions not yet completed. If no actions have been taken, the reasons shall be given.
9. **Reporting.** The acute toxicity test results shall include the contracting laboratory's complete report provided to the Permittee and shall be in accordance with section 12 (Report Preparation) of *Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms* (U.S. EPA Report No. EPA-821-R-02-012, 5th edition or subsequent editions). The submitted report shall clearly identify test results and the Permittee's status with regard to compliance with effluent limitations and other permit requirements.
10. **Ammonia Toxicity.** The acute toxicity test shall be conducted without modifications to eliminate ammonia toxicity.

B. Chronic Toxicity Testing

The Permittee shall conduct chronic toxicity testing to demonstrate compliance with the Basin Plan's water quality objective for toxicity and in accordance with chronic toxicity monitoring provisions established in section VI.C.2 of the Order. The Permittee shall meet the following chronic toxicity testing requirements:

1. **Test Frequency.** The Permittee shall conduct chronic WET testing in accordance with the schedule established by this MRP while discharging at Discharge Point 002, as summarized in Table E-4, above.
2. **Sample Type.** Effluent samples for chronic toxicity testing shall be grab samples collected at EFF-002. For toxicity tests requiring renewals, grab samples collected on consecutive days are required. When tests are conducted off-site, a minimum of three samples shall be collected, in accordance with U.S. EPA test methods.

3. **Test Species.** Test species for chronic WET testing shall be a vertebrate, the fathead minnow, *Pimephales promelas* (larval survival and growth Test Method 1000.0), an invertebrate, the water flea, *Ceriodaphnia dubia* (survival and reproduction Test Method 1002.01), and a plant, the green algae, *Selenastrum capricornutum* (also named *Raphidocelis subcapitata*) (growth Test Method 1003.0). At least one time every 5 years, the Permittee shall conduct two suites of chronic WET testing using the three species identified above. After this screening period, monitoring shall be conducted annually using the most sensitive species.
4. **Test Methods.** The presence of chronic toxicity shall be estimated as specified in U.S. EPA's *Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Water to Freshwater Organisms* (U.S. EPA Report No. EPA-821-R-02-013, or subsequent editions).

Test procedures related to pH control, sample filtration, aeration, temperature control and sample dechlorination shall be performed in accordance with the U.S. EPA method and fully explained and justified in each chronic toxicity report submitted to the Regional Water Board. The control of pH in chronic toxicity tests is allowed, provided the test pH is maintained at the pH of the receiving water measured at the time of sample collection, and the control of pH is done in a manner that has the least influence on the test water chemistry and on the toxicity of other pH sensitive materials such as some heavy metals, sulfide and cyanide.

5. **Test Dilutions.** The chronic toxicity test shall be conducted using a series of at least five dilutions and a control. The series shall consist of the following dilution series: 12.5, 25, 50, 75, and 100 percent, and a control. Effluent dilution and control water may be receiving water or standard synthetic laboratory water as described in the U.S. EPA test methods manual. Where toxicity or biostimulatory issues are not a concern in the receiving water, receiving water is preferred for control and dilution water. If the dilution water used is different from the culture water, a second control using culture water shall be used.
6. **Reference Toxicant.** If organisms are not cultured in-house, concurrent testing with a reference toxicant shall be conducted. Where organisms are cultured in-house, monthly reference toxicant testing is sufficient. Reference toxicant tests also shall be conducted using the same test conditions as the effluent toxicity tests (e.g., same test duration, etc.).
7. **Test Failure.** If either the reference toxicant test or the chronic toxicity test does not meet all test acceptability criteria, as specified in the test method, the Permittee shall re-sample and re-test as soon as possible, not to exceed 14 days following notification of test failure.
8. **Notification.** The Permittee shall notify the Regional Water Board verbally within 72 hours and in writing within 14 days after the receipt of test results exceeding the chronic toxicity monitoring trigger during regular or accelerated monitoring.
9. **Accelerated Monitoring Requirements.** If the result of any chronic toxicity test exceeds the chronic toxicity monitoring trigger of 1.6 TUC as a single sample result or 1.0 TUC as a monthly median, as specified in section VI.C.2.a.iv of the Order, and the testing meets all test acceptability criteria, the Permittee shall initiate accelerated monitoring. Accelerated monitoring shall consist of four additional effluent samples and dilution series (specified in number 5 above) – with one test for each test species showing toxicity results exceeding the toxicity trigger. Accelerated monitoring tests shall be conducted approximately every week over a four week period.

Testing shall commence within 14 days of receipt of initial sample results which indicated an exceedance of the chronic toxicity trigger. If the discharge will cease before the additional

samples can be collected, the Permittee shall contact the Executive Officer within 21 days with a plan to address elevated levels of chronic toxicity in effluent and/or receiving water. The following protocol shall be used for accelerated monitoring and TRE implementation:

- a. If the results of any accelerated toxicity testing exceed 1.0 TUc as a monthly median, the Permittee shall cease accelerated monitoring, and within 30 days of the date of completion of the accelerated monitoring, initiate the TRE Work plan developed in accordance with section VI.C.2.a.ii of the Order to investigate the cause(s) and identify actions to reduce or eliminate the chronic toxicity. Within 30 days of completing the TRE Work plan implementation, the Permittee shall submit a report to the Regional Water Board that shall include, at a minimum:
 - i. Specific actions the Permittee took to investigate and identify the cause(s) of toxicity, including a TRE WET monitoring schedule;
 - ii. Specific actions the Permittee took to mitigate the impact of the discharge and prevent the recurrence of toxicity;
 - iii. Recommendations for further actions to mitigate continued toxicity, if needed; and
 - iv. A schedule for implementation of recommended actions.
- b. If the results of four consecutive accelerated monitoring tests do not exceed the chronic toxicity trigger of 1.0 TUc, as a monthly median, the Permittee may cease accelerated monitoring and resume regular chronic toxicity monitoring. However, if there is adequate evidence of a pattern of effluent toxicity, the Regional Water Board's Executive Officer may require that the Permittee initiate a TRE.
- c. If the source(s) of the toxicity is easily identified (i.e. temporary plant upset), the Permittee shall make necessary corrections to the Facility and shall continue accelerated monitoring until four (4) consecutive accelerated tests do not exceed the monitoring trigger. Upon confirmation that the chronic toxicity has been removed, the Permittee may cease accelerated monitoring and resume regular chronic toxicity monitoring.

C. Chronic Toxicity Reporting

1. **Routine Reporting.** Chronic toxicity monitoring results shall be submitted with the monthly self-monitoring report for the month that chronic toxicity monitoring was performed. Routine reporting shall include the following in order to demonstrate compliance with permit requirements:
 - a. WET test reports shall include the contracting laboratory's complete report provided to the Permittee and shall be in accordance with the appropriate "Report Preparation and Test Review" sections of the method manuals and this MRP. The WET test report shall contain a narrative report that includes details about WET test procedures and results, including the following:
 - i. receipt and handling of the effluent sample that includes a tabular summary of initial water quality characteristics;
 - ii. the source and make-up of the lab control/diluent water used for the test;
 - iii. any manipulations done to lab control/diluent and effluent such as filtration, nutrient addition, etc.;
 - iv. identification of any reference toxicant testing performed;

- v. tabular summary of test results for control water and each effluent dilution and statistics summary to include calculation of the NOEC, TUC and IC25;
- vi. identification of any anomalies or nuances in the test procedures or results; and
- vii. summary and conclusions section.
- viii. WET test results shall include, at a minimum, for each test:
 - (a) Sample date(s);
 - (b) Test initiation date;
 - (c) Test species;
 - (d) End point values for each dilution (e.g., number of young, growth rate, percent survival);
 - (e) NOEC value(s) in percent effluent;
 - (f) IC15, IC25, IC40, and IC50 values (or EC15, EC25...etc.) in percent effluent;
 - (g) TUC values (100/NOEC);
 - (h) Mean percent mortality ($\pm$ s.d.) after 96 hours in 100 percent effluent (if applicable);
 - (i) NOEC and LOEC values for reference toxicant test(s);
 - (j) IC50 or EC50 value(s) for reference toxicant test(s);
 - (k) Available water quality measurements for each test (e.g., pH, DO, temperature, conductivity, hardness, salinity, ammonia);
 - (l) Statistical methods used to calculate endpoints;
 - (m) The statistical output page, which includes the calculation of percent minimum significant difference (PMSD); and
 - (n) Results of applicable reference toxicant data with the statistical output page identifying the species, NOEC, LOEC, type of toxicant, dilution water used, concentrations used, PMSD and dates tested; the reference toxicant control charts for each endpoint, to include summaries of reference toxicant tests performed by the contracting laboratory; and any information on deviations from standard test procedures or problems encountered in completing the test and how the problems were resolved.
- b. **Compliance Summary.** In addition to the WET report, the Permittee shall submit a compliance summary that includes an updated chronology of chronic toxicity test results expressed in NOEC and TUC for tests conducted during the permit term, and organized by test species, type of test (survival, growth or reproduction), and monitoring frequency (routine, accelerated, or TRE). Each compliance summary report shall clearly identify whether or not the effluent discharge is below the chronic toxicity monitoring triggers and, in the event that the effluent discharge exceeds a single sample or median chronic toxicity trigger, the status of efforts (e.g., accelerated monitoring, TRE, TIE, etc.) to identify the source of chronic toxicity as required by section V.B.9 of this MRP.

2. **Quality Assurance Reporting.** Because the permit requires sublethal hypothesis testing endpoints from methods 1000.0, 1002.0, and 1003.0 in the test methods manual titled *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (U.S. EPA Report No. EPA-821-R-02-013, 2002, or subsequent editions), with-in test variability must be reviewed for acceptability and variability criteria (upper and lower PMSD bounds) must be applied, as directed under section 10.2.8 – *Test Variability* of the test methods manual. Under section 10.2.8, the calculated PMSD for both reference toxicant test and effluent toxicity test results must be compared with the upper and lower PMSD bounds variability criteria specified in Table 6 – *Variability Criteria (Upper and Lower PMSD Bounds) for Sublethal Hypothesis Testing Endpoints Submitted Under NPDES Permits*, following the review criteria in paragraphs 10.2.8.2.1 through 10.2.8.2.5 of the test methods manual. Based on this review, only accepted effluent toxicity test results shall be reported.

VI. LAND DISCHARGE MONITORING REQUIREMENTS

A. Monitoring Location LND-001

1. The Permittee shall monitor treated wastewater at Monitoring Location LND-001 as follows:

Table E-5. Land Discharge Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow ¹	mgd	Meter	Continuous	Meter
pH	Standard units	Grab	Daily	Standard Methods ²
Nitrate (as N)	mg/L	Grab	Monthly	Standard Methods ²
Total Dissolved Solids	mg/L	Grab	Monthly	Standard Method 2540C
Sodium	mg/L	Grab	Monthly	Standard Methods ²
Aluminum	mg/L	Grab	Monthly	Standard Methods ²
Visual Observations ³	---	---	Daily	Visual
Table Notes: 1. Each month, the Permittee shall report the number of days that treated wastewater was used for irrigation on the Burch property, as well as the average and maximum daily flow rate to each property. 2. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or current test procedures specified in 40 C.F.R. Part 136. 3. Visual observations shall be conducted during and immediately after any discharge to the irrigation system, and shall include a record of any odors, evidence of surface run-off, or other signs of malfunction or improper operation. The monthly monitoring report shall include the daily volume of treated wastewater distributed to the irrigation field, and any observations indicating non-compliance with the provisions of the waste discharge requirements.				

VII. RECYCLING MONITORING REQUIREMENTS

A. Monitoring Location REC-001

1. The Permittee shall monitor recycled water distributed for irrigation at Monitoring Location REC-001 as follows:

Table E-6a. Recycling Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow ¹	mgd	Meter	Continuous	Meter
Nitrate Nitrogen, Total (as N) ²	mg/L	Grab	Monthly	Standard Methods ³
Nitrite Nitrogen, Total (as N) ²	mg/L	Grab	Monthly	Standard Methods ³
Ammonia Nitrogen, Total (as N) ²	mg/L	Grab	Monthly	Standard Methods ³
Organic Nitrogen, Total (as N) ²	mg/L	Grab	Monthly	Standard Methods ³
Total Dissolved Solids (TDS)	mg/L	Grab	Monthly ⁴	Standard Methods ³
Chloride	mg/L	Grab	Monthly ⁴	Standard Methods ³
Boron	mg/L	Grab	Monthly ⁴	Standard Methods ³
Sodium	mg/L	Grab	Monthly ⁴	Standard Methods ³
Visual Observations ⁵	--	--	---	Visual
Table Notes: - Each month, the Permittee shall report the number of days that treated wastewater was used for recycled water irrigation at the Northwood Golf Course or other approved recycled water use sites, as well as the average and maximum daily flow rate. - Monitoring for nitrate, nitrite, ammonia and organic nitrogen is for the purpose of determining total nitrogen concentration for agronomic rate calculations. - In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or current test procedures specified in 40 C.F.R. Part 136. - The monitoring frequency for TDS, chloride, boron, and sodium may be reduced to annually if the previous year of monitoring data demonstrates that concentrations of these constituents are consistently lower than water quality objectives for protection of groundwater. - During periods of discharge to the irrigation system, visual observations shall be conducted at least monthly for agronomic applications to verify compliance with recycled water requirements in Attachment G. The inspection frequency shall be increased for use sites with a history of non-compliance with water recycling requirements established in this Order. Visual monitoring shall confirm proper operation of the recycled water system and associated BMPs. The Permittee shall include a record of any malfunctions or findings of improper operation, including, but not limited to odors, evidence of surface run-off, or ponding that exceeds 24-hours. Visual observations may be performed by the irrigation users in accordance with the Permittee's user agreements. The monthly recycled water report shall include the daily volume of treated wastewater discharged to the irrigation system and any observations indicating non-compliance with the provisions of the waste discharge requirements.				

- The Permittee shall comply with Water Recycling Requirements and Provisions contained in Attachment G of this Order.

B. Recycled Water Production and Use

- Recycled water quality characteristics and precipitation data shall be used to ascertain nitrogen loading rates at each recycled water use site. The following information shall be reported for each use site or use site type.
- Reporting of the required monitoring shall be provided annually in the Annual Recycled Water Report to the Regional Water Board, and annually to recycled water users.

Table E-6b. Recycled Water Production and Use

Parameter	Units	Sample Type	Minimum Monitoring Frequency
Volume of recycled water ¹	Acre-feet	Meter	Monthly
Total area of application	Acres	Observation	Monthly
Total Nitrogen application rate ^{2,3}	lbs/acre-month	Calculation	Monthly
Rainfall	Inches	Gage	Daily
Table Notes: 1. Estimation of the volume of recycled water shall not include other potable or non-potable "make-up" water used in conjunction with recycled water. 2. Nitrogen application rate shall consider nitrogen content of the recycled water, based on effluent monitoring data. 3. Nitrogen concentrations shall be calculated and reported "as N". For example, nitrate-nitrogen = 27 mg/L as NO ₃ shall be converted and reported as nitrate-nitrogen = 6.1 mg/L as N using a conversion factor of 14.067 (N)/62.0049 (NO ₃).			

VIII. RECEIVING WATER MONITORING REQUIREMENTS

A. Surface Water Monitoring Locations RSW-001 and RSW-002

- The Permittee shall monitor upstream and downstream conditions in the Russian River during the discharge season at Monitoring Locations RSW-001 and RSW-002 as follows:

Table E-7a. Receiving Water Monitoring Requirements – RSW-001 and RSW-002

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow ¹	mgd	Meter	Daily	---
pH	pH units	Grab	Weekly	Standard Methods ²
Dissolved Oxygen	mg/L	Grab	Weekly	Standard Methods ²
Temperature	°C	Grab	Weekly	Standard Methods ²
Turbidity	NTU	Grab	Weekly	Standard Methods ²
Specific Conductance ³	µmhos/cm	Grab	Monthly	Standard Methods ²
Total Dissolved Solids	mg/L	Grab	Monthly	Standard Methods ²
Aluminum	mg/L	Grab	Monthly	Standard Methods ²
Ammonia Nitrogen, Total (as N)	mg/L	Grab	Monthly	Standard Methods ²
Unionized Ammonia	mg/L		Monthly	Calculation
Nitrate Nitrogen, Total (as N)	mg/L	Grab	Monthly	Standard Methods ²
Organic Nitrogen, Total (as N)	mg/L	Grab	Monthly	Standard Methods ²
Total Phosphorus, Total (as P)	mg/L	Grab	Monthly	Standard Methods ²
CTR Priority Pollutants ^{4, 5, 6}	µg/L ⁷	Grab	Once per Permit Term	40 C.F.R. 136
Hardness (CaCO ₃)	mg/L	Grab	Concurrent with priority pollutant sampling	Standard Methods ²
Table Notes: 1. The flow rate shall be determined using USGS Gage No. 11-4670.00 at the Hacienda Bridge, and compared to the daily discharge rate to determine compliance with Discharge Prohibition III.K of the Order. For each month during the discharge season, peak daily and average daily flow shall be reported. 2. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or current test procedures specified in 40 C.F.R. Part 136. 3. Measured in micromhos/cm at 25° C.				

4. Those pollutants identified as Compound Nos. 1 – 126 by the California Toxics Rule at 40 C.F.R. 131.38(b)(1). Samples shall be collected on the same day as receiving water samples are collected for analysis of the priority pollutants. Analyses for the priority pollutants shall be conducted in accordance to methods established at 40 C.F.R. Part 136, or if no method is specified for a pollutant at 40 C.F.R. Part 136, in accordance to methods approved by the State Water Resources Control Board or the Regional Water Board.
5. Monitoring for CTR priority pollutants is only required at RSW-001.
6. Analytical methods shall achieve the minimum levels (ML) specified in Appendix 4 of the SIP. If no ML value is below the effluent limitation, the lowest ML shall be selected as the reporting level (RL). In accordance with section 2.4.1 of the SIP, the Permittee shall report the Reporting Level (RL) and the Method Detection Limit (MDL) with each sample result.
7. Or other units as appropriate.

B. Groundwater Monitoring Locations GW-001, GW-002, and GW-003

1. The Permittee shall monitor groundwater to assess impacts of land discharge in the lower Burch disposal area at Monitoring Locations GW-001, GW-002, and GW-003 as follows:

Table E-7b. Receiving Water Monitoring Requirements - Groundwater

Parameter	Units	Sample Type	Minimum Sampling Frequency ¹	Required Analytical Test Method
pH	Standard units	Grab	Monthly	Standard Methods ²
Nitrate (as N)	mg/L	Grab	Monthly	Standard Methods ²
Total Dissolved Solids	mg/L	Grab	Monthly	Standard Methods ²
Sodium	mg/L	Grab	Monthly	Standard Methods ²
Chloride	mg/L	Grab	Monthly	Standard Methods ²
Aluminum	mg/L	Grab	Monthly	Standard Methods ²
Depth to Groundwater	inches	Grab	Monthly	Measurement
Table Notes:				
1. The groundwater monitoring frequency may be reduced to quarterly, upon demonstration by the Permittee that the disposal of wastewater has not caused groundwater quality objectives to be exceeded or statistically significant degradation of groundwater quality, for a period of two years, and approval by the Regional Water Board Executive Officer.				
2. In accordance with the current edition of Standard Methods for Examination of Water and Wastewater (American Public Health Administration) or current test procedures specified in 40 C.F.R. Part 136.				

IX. OTHER MONITORING REQUIREMENTS

A. Filtration Process Monitoring

Filtration process monitoring shall demonstrate compliance with section IV.D.1 (Filtration Process Requirements) of the Order and applies to all treated wastewater flows. The Permittee is required to implement the following filtration process monitoring:

1. Effluent Filter Monitoring (Monitoring Location INT-001A)

- a. **Monitoring.** The Permittee shall calculate on a daily basis the surface loading rate in gallons per minute per square feet and report the maximum surface loading rate and any exceedances of the surface loading rate limitations specified in section IV.D.1.a of the Order. The rate of flow through the tertiary filters shall be measured at Monitoring Location INT-001A.
- b. **Compliance.** Compliance with the maximum daily filter surface loading rate as specified in section 60301.320 of title 22, shall be calculated based on the flow rate through each filter unit.

- c. **Reporting.** The maximum daily filter surface loading rate shall be reported on the monthly self-monitoring report.

2. Effluent Filter Monitoring (Monitoring Location INT-001B)

- a. **Monitoring.** The turbidity of the filter effluent shall be continuously measured and recorded at Monitoring Location INT-001B. Should the turbidity meter and recorder fail, grab sampling at a minimum frequency of 1.2 hours may be substituted for a period of up to 24 hours. The recorded data shall be maintained by the Permittee for at least 3 years. The daily maximum and 95th percentile turbidity results shall be reported on the monthly monitoring reports.
- b. **Compliance.** Compliance with the effluent turbidity limitation specified in section IV.D.1.b. (Filtration Process Requirements) of this Order shall be determined using the levels of recorded turbidity taken at intervals of no more than 1.2 hours over a 24-hour period.
- c. **Reporting.** If the filter effluent turbidity exceeds an average of 2 NTU during a 24-hour period, 5 NTU more than 5 percent of the time during a 24-hour period, or 10 NTU at any time, the incident shall be reported in the monthly self-monitoring report and the incident shall be reported to the Regional Water Board and CDPH by telephone within 24 hours in accordance with Provision VI.A.2.b of the Order only if the effluent was sent to the recycled water distribution system or discharged to the Russian River. A written report describing the incident and the actions undertaken in response shall be included in the monthly self-monitoring report. Mitigation of the event shall consist of diverting the non-compliant effluent, land application at Discharge Point 003, or automatically activated chemical addition to comply with title 22 requirements (sections 60304 and 60307).

B. Disinfection Process Monitoring for UV Disinfection System (Monitoring Location INT-002)

- 1. **Monitoring.** The UV transmittance of the effluent from the UV disinfection system shall be monitored continuously and recorded at Monitoring Location INT-002. The operational UV dose shall be calculated from UV transmittance and flow.
- 2. **Compliance.** The UV transmittance shall not fall below 55 percent of maximum at any time, unless otherwise approved by CDPH. The operational UV dose shall not fall below 100 millijoules per square centimeter (mJ/cm^2) at any time, unless otherwise approved by CDPH. Flow through the UV disinfection system shall not exceed 3.7 mgd, unless otherwise approved by CDPH.
- 3. **Reporting.** The Permittee shall report daily average and lowest daily transmittance and operational UV dose on its monthly monitoring reports. The Permittee shall report daily average and minimum flow through the UV disinfection system. If the UV transmittance falls below 55 percent or UV dose falls below $100 \text{ mJ}/\text{cm}^2$, the event shall be reported to the Regional Water Board and the CDPH by telephone within 24 hours. Any inadequately treated and disinfected wastewater shall be diverted to a storage basin or an upstream process for adequate treatment.

C. Visual Monitoring at Monitoring Locations LND-001 and REC-001

Visual observations of the discharge and the receiving water shall be recorded monthly and on the first day of each intermittent discharge. Visual monitoring shall include, but not be limited to, observations for floating materials, coloration, objectionable aquatic growths, oil and grease

films, and odors. Visual observations shall be recorded and included in the Permittee's monthly monitoring reports.

D. Sludge Monitoring (Monitoring Location BIO-001)

1. Sludge sampling shall be conducted according to the requirements specified by the location and type of disposal activities undertaken.
2. Sampling records shall be retained for a minimum of 5 years. A log shall be maintained for sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary, however, the log must be complete enough to serve as a basis for developing the Sludge Handling and Disposal report that is required as part of the Annual Report.

X. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

1. The Permittee shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.

B. Self-Monitoring Reports (SMRs)

1. The Permittee shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). The CIWQS Web site will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal.
2. The Permittee shall maintain sufficient staffing and resources to ensure it submits eSMRs that are complete and timely. This includes provision of training and supervision of individuals (e.g., Permittee personnel or consultant) on how to prepare and submit eSMRs.
3. The Permittee shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. The Permittee shall submit monthly, quarterly, and annual SMRs including the results of all required monitoring using U.S. EPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Permittee monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
4. All monitoring results reported shall be supported by the inclusion of the complete analytical report from the laboratory that conducted the analyses.
5. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table E-8 Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Continuous	Permit effective date	All	First day of second calendar month following month of sampling
Daily	Permit effective date	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	First day of second calendar month following month of sampling

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	First day of second calendar month following month of sampling
Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1 st day of calendar month through last day of calendar month	First day of second calendar month following month of sampling
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit effective date	January 1 through March 31 April 1 through June 30 July 1 through September 30 October 1 through December 31	First day of second calendar month following end of quarter
Semiannually	Closest of January 1 or July 1 following (or on) permit effective date	January 1 through June 30 July 1 through December 31	March 1 of each year (with annual report)
Annually	January 1 following (or on) permit effective date	January 1 through December 31	March 1 of each year (Annual Report)
Once per Discharge Season	Permit effective date	October 1 through May 14, during a period of discharge to surface waters	July 1 of each year
Once per Permit Term	Permit effective date	October 1 through May 14, during a period of discharge to surface waters	With application for permit renewal

- 6. Reporting Protocols.** The Permittee shall report with each sample result the applicable Reporting Level (RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 C.F.R. part 136.

The Permittee shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- c. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
- d. Permittees are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to

calibration standards) is the lowest calibration standard. At no time is the Permittee to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.

7. **Compliance Determination.** Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined above. For purposes of reporting and administrative enforcement by the Regional Water Board and State Water Board, the Permittee shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).
8. **Multiple Sample Data.** When determining compliance with an AMEL or MDEL for priority pollutants and more than one sample result is available, the Permittee shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Permittee shall compute the median in place of the arithmetic mean in accordance with the following procedure:
 - a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
9. The Permittee shall submit SMRs in accordance with the following requirements:
 - a. The Permittee shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the Facility is operating in compliance with interim and/or final effluent limitations. The reported data shall include calculation of all effluent limitations that require averaging, taking of a median, or other computation. The Permittee is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Permittee shall electronically submit the data in a tabular format as an attachment. The Permittee's reports shall clearly identify the Discharge or Distribution Points that were utilized during the monitoring period. During periods when there is no discharge to one more Discharge or Distribution Points, the reports shall certify "No Discharge".
 - b. The Permittee shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify:
 - i. Facility name and address;
 - ii. WDID number;
 - iii. Applicable period of monitoring and reporting;
 - iv. Violations of the WDRs, including a description of any requirement not complied with and a description of the event, and the reason for the violation;

- v. Corrective actions taken or planned; and
- vi. The proposed time schedule for corrective actions.
- c. SMRs must be submitted to the Regional Water Board, signed and certified as required by the Standard Provisions (Attachment D), to the CIWQS Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). In the event that paper submittal of SMRs is required, the Permittee shall submit the SMR to the address listed below:

Regional Water Quality Control Board
 North Coast Region
 5550 Skylane Blvd., Suite A
 Santa Rosa, CA 95403

C. Discharge Monitoring Reports (DMRs)

- 1. At any time during the term of this permit, the State Water Board or Regional Water Board may notify the Permittee to electronically submit DMRs. Until such notification is given specifically for the submittal of DMRs, the Permittee shall submit DMRs in accordance with the requirements described below.
- 2. DMRs must be signed and certified as required by the standard provisions (Attachment D). The Permittee shall submit the original DMR and one copy of the DMR to the address listed below:

STANDARD MAIL	FEDEX/UPS/ OTHER PRIVATE CARRIERS
State Water Resources Control Board Division of Water Quality c/o DMR Processing Center PO Box 100 Sacramento, CA 95812-1000	State Water Resources Control Board Division of Water Quality c/o DMR Processing Center 1001 I Street, 15 th Floor Sacramento, CA 95814

Permittees operating a “minor” facility, if so designated in the fact sheet, are excepted from submitting DMRs under these requirements. However, at any time during the term of this permit, the State Water Board or Regional Water Board may notify such a Permittee to submit DMRs, at which time this exception will no longer apply.

- 3. All discharge monitoring results must be reported on the official U.S. EPA pre-printed DMR forms (EPA Form 3320-1) or on self-generated forms that follow the exact same format of EPA Form 3320-1.

D. Other Reports

- 1. The Permittee shall report the results of any special studies, acute and chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special Provisions – VI.B and VI.C. of the Order. The Permittee shall report the progress in satisfaction of compliance schedule dates specified in Special Provisions – VI.C.7. The Permittee shall submit reports with the first monthly SMR scheduled to be submitted on or immediately following the report due date in compliance with SMR reporting requirements described in subsection X.B above.
- 2. The Permittee shall report progress in satisfaction of compliance schedule dates, if any, specified in Special Provision – VI.C.7 of this Order. Progress reports shall be submitted on,

or before, each compliance due date, and shall identify compliance or noncompliance with the specific date and task. If noncompliance is reported, the Permittee shall state the reasons for noncompliance and include an estimate of the date when the Permittee will be in compliance. The Permittee shall notify the Regional Water Board by letter when it returns to compliance with the compliance time schedule.

3. Water Recycling System

- a. Water Recycling Operations Reporting.** The Permittee shall submit reports pertaining to the operation, performance, monitoring, and other activities related to water recycling as follows:
 - i. Monthly Recycled Water Report.** The Permittee shall submit a monthly recycled water summary report, as required by section 13523.1(b)(4) of the Water Code and other requirements of this Order, containing the following information:
 - (a)** Total volume of recycled water supplied to each recycled water user for each month of the reporting period;
 - (b)** Total number of recycled water use sites;
 - (c)** Locations of recycled water use sites, including a map and tabular summary with acreage and name of property owner;
 - (d)** A summary of recycled water use site inspections conducted by the Permittee or recycled water users. Required reporting includes the number and dates of inspections conducted for each use site during the reporting period; all observations of recycled water over-application and/or runoff; and the number of observations of noncompliance for each use site including description of the noncompliance and its cause, the period of noncompliance, and if the noncompliance has not been corrected, the anticipated time it is expected to continue and steps taken or planned to reduce, eliminate, and prevent recurrence of the noncompliance.
 - (e)** A summary of operational problems, plant equipment malfunctions, and any diversion of recycled water which does not meet the requirements specified in this Order.
 - (f)** Documentation of notifications to users if any recycled water was delivered that did not meet the requirements specified in this Order.
 - (g)** A record of equipment or process failures initiating an alarm that prevented recycled water from meeting the requirements of this Order, as well as any corrective and preventative actions.
 - ii. Annual Recycled Water Report.** The annual report shall include but not be limited to the following:
 - (a)** A compliance summary and discussion of the compliance record for the prior calendar year, including:
 - (1)** In the event of noncompliance, the report shall also discuss the corrective actions taken and planned to bring the recycled water program into full compliance with this Order;

- (2) Certification that all reasonable BMPs and management practices were implemented to ensure efficient and compliant operation of the recycled water system; and
 - (3) Identification of any other problems that occurred in the recycled water system during the prior year, including repeated occurrences of incidental runoff of which the Permittee is aware, and plans to rectify those problems in the coming year.
 - (b) A summary of major repairs scheduled or completed that affected the recycled water system appurtenances and irrigation areas;
 - (c) Enforcement and monitoring activities that occurred during the previous year, and identification of any problems and how the problems were addressed;
 - (d) If applicable, a summary of all cross-connection testing and back-flow prevention activities (inspections, maintenance) and a summary of any problems identified, or certification that no problems occurred;
 - (e) Documentation of compliance with California Health and Safety Code section 116815 as specified in Water Recycling Requirement B.17 of Attachment G regarding the installation and marking of new sites or newly retrofitted recycled water piping; and
 - (f) A description of agronomic rate compliance, pursuant to section VII.B of the MRP.
- 4. **Annual Report.** The Permittee shall submit an annual report to the Regional Water Board for each calendar year electronically through the CIWQS Program web site. In the event that a paper copy of the annual report is required, the Permittee shall submit the report to the address in section X.B.9.c., above. The report shall be submitted by March 1 of the following year. The report shall, at a minimum, include the following:
 - a. Both tabular and, where appropriate, graphical summaries of the monitoring data and disposal records from the previous year. If the Permittee monitors any pollutant more frequently than required by this Order, using test procedures approved under 40 C.F.R. section 136 or as specified in this Order, the results of this monitoring shall be included in the calculation and report of the data submitted SMR.
 - b. A comprehensive discussion of the Facility's compliance (or lack thereof) with all effluent limitations and other WDRs, and the corrective actions taken or planned, which may be needed to bring the discharge into full compliance with the Order.
 - c. The names and telephone numbers of persons to contact regarding the wastewater treatment Facility for emergency and routine situations.
 - d. A statement certifying when the flow meter(s) and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration.
 - e. A statement certifying whether the current operation and management manual and spill contingency plan, reflect the wastewater treatment Facility as currently constructed and operated, and the dates when these documents were last reviewed and last revised for adequacy.

- f. Source Control Activity Reporting.** The Permittee shall submit, as part of its annual report to the Regional Water Board, a description of the Permittee's source control activities over the previous 12-month period, as required by Provision VI.C.5.b. of this Order.
- i.** A copy of the source control standards.
 - ii.** A description of the waste hauler permit system, if applicable.
 - iii.** A summary of the compliance and enforcement activities taken by the Permittee during the past year, which ensures industrial user compliance. The summary shall include the names and addresses of any industrial or commercial users under surveillance by the Permittee, an explanation of whether they were inspected, sampled, or both, the frequency of these activities at each user, and the conclusions or results from the inspection or sampling of each user.
 - iv.** An updated list of industrial users (by North American Industrial Classification/Standard Industrial Classification categories) which were issued permits and/or enforcement orders, and a status of compliance for each user.
 - v.** The name and address of each user that received a discharge limit.
 - vi.** A summary of any industrial waste survey results.
 - vii.** A summary of public outreach activities conducted by the Permittee to educate industrial, commercial, and residential users about preventing discharges of industrial and toxic wastes to the wastewater treatment plant.
- g. Sludge Handling and Disposal Activity Reporting.** The Permittee shall submit, as part of its annual report to the Regional Water Board, a description of the Permittee's solids handling, disposal and reuse activities over the previous twelve months. At a minimum, the report shall contain:
- i.** Annual sludge production, in dry tons and percent solids;
 - ii.** Sludge monitoring results; and
 - iii.** A schematic diagram showing sludge handling facilities (e.g., digesters, thickeners, drying beds, etc.), if any, and a solids flow diagram.
 - iv.** Methods of final disposal of sludge:
 - (a)** For any portion of sludge discharged to a sanitary landfill, the Permittee shall provide the volume of sludge transported to the land fill, the names and locations of the facilities receiving sludge, the Regional Water Board's WDRs order number for the regulated landfill, and the landfill classification.
 - (b)** For any portion of sludge discharged through land application, the Permittee shall provide the volume of biosolids applied, the date and locations where biosolids were applied, the Regional Water Board's WDRs order number for the regulated discharge, a demonstration that the discharge was conducted in compliance with applicable permits and regulations, and, if applicable, corrective actions taken or planned to bring the discharge into compliance with WDRs.

- (c) For any portion of sludge further treated through composting, the Permittee shall provide a summary of the composting process, the volume of sludge composted, and a demonstration and signed certification statement that the composting process and final product met all requirements for Class A biosolids.
 - (d) Results of internal or external third-party audits of the Biosolids Management System, including reported program deficiencies and recommendations, required corrective actions, and a schedule to complete corrective action.
- h. **Storm Water Reporting.** The Permittee shall submit, as part of its annual report to the Regional Water Board, an evaluation of the effectiveness of the Permittee's BMPs to control the run-on of storm water to the treatment facility site, as well as activities to maintain and upgrade these BMPs.
- i. **Flood Control and Flow Reduction Mitigation Reporting.** The Permittee shall submit, as part of its annual report to the Regional Water Board, a summary of all flood control and flow reduction mitigation measures that the Permittee implemented in the prior year and provide an evaluation of the effectiveness of those flood control and flow reduction mitigation measures and recommendations for improving the flood control and flow reduction mitigation program for the upcoming year.

E. Spill Notification

- 1. Spills and Unauthorized Discharges.** Information regarding all spills and unauthorized discharges (except SSOs and recycled water) that may endanger health or the environment shall be provided orally to the Regional Water Board¹ within 24 hours from the time the Permittee becomes aware of the circumstances and a written report shall also be submitted within five (5) days of the time the Permittee becomes aware of the circumstances, in accordance with Section V.E. of Attachment D.

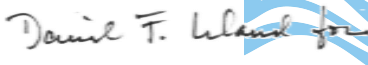
Information to be provided verbally to the Regional Water Board includes:

- a. Name and contact information of caller;
 - b. Date, time and location of spill occurrence;
 - c. Estimates of spill volume, rate of flow, and spill duration, if available and reasonably accurate;
 - d. Surface water bodies impacted, if any;
 - e. Cause of spill, if known at the time of the notification;
 - f. Cleanup actions taken or repairs made at the time of the notification; and
 - g. Responding agencies.
2. **Sanitary Sewer Overflows.** Notification and reporting of sanitary sewer overflows is conducted by the Permittee in accordance with the requirements of Order No. 2006-0003-DWQ (Statewide General WDRs for Sanitary Sewer Systems) and any revisions thereto.
3. **Recycled Water Spills.** Notification and reporting of spills and unauthorized discharges of recycled water discharged in or on any waters of the State, as defined in Water Code section 13050, shall be conducted in accordance with the following:

¹ The contact number of the Regional Water Board during normal business hours is (707) 576-2220. After normal business hours, spill reporting to CalEMA will satisfy the 24 hour spill reporting requirement for the Regional Water Board. The contact number for spill reporting for the CalEMA is (800) 852-7550.

a. Tertiary Recycled Water²

- i. For unauthorized discharges of 50,000 gallons or more of tertiary recycled water, the Permittee shall immediately notify the Regional Water Board as soon as (a) the Permittee has knowledge of the discharge or probable discharge; (b) notification is possible; and (c) notification can be provided without substantially impeding cleanup or other emergency measures.
- ii. For unauthorized discharges of more than 1,000 gallons, but less than 50,000 gallons of tertiary recycled water, the Permittee shall notify the Regional Water Board as soon as possible, but no longer than three days after becoming aware of the discharge.

 Digitally signed by David F.
Leland
Date: 2015.05.19 12:10:23
07'00'

Ordered by: _____

Matthias St. John
Executive Officer

May 19, 2015

14_0002_RRCSD_NPDES_Permit_MRP_Revision

² Tertiary Recycled Water means “disinfected tertiary 2.2 recycled water” as defined by CDPH or wastewater receiving advanced treatment beyond disinfected tertiary 2.2 recycled water.